

Yubin (Olivia) Zhang

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Education

University of Toronto

Sept. 2022 – May 2027

Bachelor of Applied Science and Engineering with PEY-COOP in Electrical and Computer Engineering

Toronto, ON

- **Minor:** Artificial Intelligence Engineering, Engineering Business
- **Relevant Course:** Operating System, Algorithm & Data Structure, Deep Learning, Computer Architecture, Data Analysis

Skills

Programming: Python, Java, C, C++, PyTorch, Verilog, HTML/CSS, Siebel CRM, JavaScript, SQL
Developer Tools: Linux, Git, Azure DevOps, Docker, React, VS Code, NetBeans, Jupyter Notebook
Technologies/Frameworks: Numpy, Pandas, Matplotlib, Tensorflow, Spark, V4L2, Spring Boot
Data & Cloud: Power BI, Excel, Microsoft Azure (App Service, Container Apps, CI/CD), Amazon Web Services
Collaborative Skills: Collaboration, Communication, Leadership, Active Learning, Microsoft 365

Experience

Programmer Co-op

May. 2025 – Present

Ontario Public Service – Enterprise Architecture Office

Toronto, ON

- Engineered and deployed **3+ full-stack government applications** following standard **SDLC** and agile sprint practices.
- Led a team of **4 developers** to deliver AI-powered roadmap tools adopted by **20+ stakeholders** across multiple ministries.
- Built **Python/Flask APIs** with **MongoDB**, supporting **100+ roadmap records** and enterprise workflow automation.
- Improved internal Q&A accuracy by **70%** via optimized **RAG pipelines** using **Azure OpenAI** and citation-backed grounding.
- Migrated services from **Azure App Service** to **Container Apps**, improving scalability and reducing deployment time by **40%**.
- Resolved **30+ backend/integration issues** through **debugging and root-cause analysis**, reducing recurring errors by **25%**.

Software Development Engineer Intern

May 2024 – Sept. 2024

MOTORCOMM ELECTRONIC TECHNOLOGY CO.LTD

Shanghai, China

- Built a **C++ V4L2** video capture app on **NVIDIA Jetson Nano** streaming **RAW12** camera sensor data at **30 FPS**.
- Implemented a **CUDA-accelerated Bayer (RGGB12)** to RGB demosaicing pipeline, reducing per-frame processing latency from **45ms to 15ms (3x speedup)** compared to CPU-only conversion.
- Optimized GPU memory transfers to sustain **real-time RGB output** with **95%+ frame stability** under continuous streaming.
- Integrated the module into an imaging pipeline with **5 engineers** for **client hardware validation** and automated sensor testing.

Investment Analysis Intern

May 2023 - Sept. 2023

Chobe Capital

Shanghai, China

- Analyzed **10+ industry sectors** using **Power BI, Excel, and SQL**, generating weekly market trend reports for senior analysts.
- Built financial models and dashboards tracking **50+ equities**, supporting portfolio evaluation and investment strategy discussions.
- Delivered data-driven insights improving **investment decision-making** and reporting accuracy for internal reviews.
- Automated repetitive analysis workflows with **Python**, reducing manual processing time by **30%** and saving **5+ hrs/week**.

Projects

Traveller Guided Map | C++, Searching Algorithm, Git, GTK, UI/UX Design

Jan. 2024 – Apr. 2024

- Developed a GIS mapping tool in **C++** and **GTK**, visualizing **60k+ map intersections** in city-scale datasets.
- Implemented **A* routing** and optimization algorithms (**Greedy, Simulated Annealing**), improving computation speed by **35%**

DE1-SoC Computer Hardware System Design | C, FPGA, Verilog, VGA, FSM

Jan. 2024 – Apr. 2024

- Designed and implemented an interactive Sokoban-style game on **DE1-SoC FPGA** using **C** and low-level hardware control.
- Integrated a **VGA display** with **double buffering** to deliver smooth real-time animations at **60 FPS** with audio feedback and switch-based difficulty level selection.
- Developed complete game logic using **Finite State Machines (FSM)**, including robot movement, collision checking, and win/loss conditions for an engaging user experience (**UX**).

Age Estimation System 🤖 | Python, Deep Learning, Machine Learning, Artificial Intelligence

May 2023 – Sept. 2023

- Developed a face-based age classification system in **Python** to distinguish adults vs. minors using deep learning.
- Preprocessed **3000+** facial images using **PyTorch** and trained a **CNN** baseline model for binary classification.
- Optimized the model using **VGG19 transfer learning** on a two-layer fully connected classifier, achieving **96.49%** accuracy in **training**, **87.28%** in **validation**, and **86.45%** in **testing**, outperforming the baseline by **2%**.